Investigating the Power Budget of a 3U Nanosatellite Designed for Earth Observation

Amina DAGHOURI
Energy Optimization, Diagnosis and Control Team, ENSAM
University Center for Research in Space Technologies, EMI
Mohammed V University in Rabat, Morocco
amina_daghouri@um5.ac.ma

Youssef EL HACHIMI
Power Electronics, Smart Control Techniques, Automatic and Robotics Team, EMI
University Center for Research in Space Technologies, EMI
Mohammed V University in Rabat, Morocco
youssef.elhachimi@research.emi.ac.ma

Abdelaali OUHAMMAM
Power Electronics, Smart Control Techniques, Automatic and Robotics Team, EMI
University Center for Research in Space Technologies, EMI
Mohammed V University in Rabat, Morocco
abdelaali_ouhammam@um5.ac.ma

Mohammed Alae CHANOUI
LASTIMI Laboratory, EST
University Center for Research in Space Technologies, EMI
Mohammed V University in Rabat, Morocco
mohammedalae_chanoui@um5.ac.ma

Soumia ELHANI
Energy Optimization, Diagnosis and Control Team, ENSAM
University Center for Research in Space Technologies, EMI
Mohammed V University in Rabat, Morocco
s.elhani@um5r.ac.ma

Hassane MAHMOUDI
Power Electronics, Smart Control Techniques, Automatic and Robotics Team, EMI
University Center for Research in Space Technologies, EMI
Mohammed V University in Rabat, Morocco
mahmoudi@emi.ac.ma

2023 IEEE 10th International Workshop on Metrology for AeroSpace (MetroAeroSpace) | 978-1-6654-5690-6/23/$31.00 ©2023 IEEE | DOI: 10.1109/METROAEROSPACE57412.2023.10190016

***Abstract*—The Electrical Power System (EPS) is one of the most critical subsystems of a nanosatellite. Thus, its design should be done carefully for a successful mission. To verify that the power needs of its subsystems does not exceed the power available, a power budget should be established. The purpose of this work is to evaluate the 3U nanosatellite's EPS by analyzing its power budget using Systems Tool Kit (STK) software. In this study, the orbital design, the sizing of the PV panels and the batteries used are presented. The simulation is done taking into account the worst case assumptions: the sun-synchronous Low Earth Orbit (LEO) orbit with the longest eclipse period, the worst case solar irradiance of 1316 W/m^2. The results of the power budget established confirm the selection and the sizing of PV panels and the batteries.**

***Index Terms*—CubeSat, Nanosatellite, EPS, Power consumption, Power budget, Earth observation.**

I. INTRODUCTION

The class of nanosatellites known as "CubeSats" currently provides a low-cost satellite that can be used for technology demonstrations, scientific research, etc. A 1 U CubeSat can be defined as a cube of maximum mass of 1.33 kg and a size of 1000 cm^3 [1].

Generally, CubeSats are launched into Low Earth Orbit (LEO) that is defined by an altitude of 250 km to 2000 km and an orbital period of 90 to 120 min [2]. Their inclination angle can range from 0 to 98° with regard to the equatorial plane. LEO can be divided into several types based on inclination and also depending on the mission's type.

The sun-synchronous orbit (SSO), which is a polar orbit inclined, is the most convenient for earth observation and mapping missions because the spacecraft sees a point on Earth with the same illumination and the same time each day.

For such a mission, SSO usually has a sunlight period of 2/3 of the orbit period and an eclipse of 1/3 of the orbit period. The eclipse period depends on the β angle between the sun vector and the orbit plane, which vary with seasons and orbit perturbations. As β increases, the eclipse period decreases until there is no longer a shadow [3].

A nanosatellite is made up of several subsystems, including the Command and Data Handling (CDH), Communication, Thermal Control, Guidance, Navigation and Control (GNC), and Electrical Power Subsystem (EPS) [4]. The EPS is one of the most important subsystems. It is responsible for managing and distributing the power buses that supply the subsystems [5], as well as generating power from PV panels and supplying regulated power to charge the spacecraft's batteries [6], allowing it to continue operating during eclipses. It is critical to design a reliable EPS because there is no way to maintain it after launch.

When dealing with the EPS design, and especially the power storage system, there is an interest in reducing the duration of the eclipse but the constraints of the mission nature should be

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on March 11,2024 at 21:33:21 UTC from IEEE Xplore. Restrictions apply.

considered. The power budget for a nanosatellite is a critical aspect of the design and operation of the satellite, and careful attention must be paid to ensure that the power needs of the various systems on board can be met within the constraints of the available power.

Several studies have explored power budget analysis and the challenges in designing CubeSats' Electrical Power System (EPS) validation. Authors in [7], examine a 1U CubeSat's power budget in a LEO polar orbit, discussing power consumption for different subsystems and power sources such as solar panels and batteries. They use a power analysis tool to estimate power consumption and recommend optimizing the battery capacity for a successful mission duration. The methodology described in this study has been applied to several 1U CubeSats such as SwissCube [8], IGOsat [9] and OUFTI [10] to verify the space segment's operational mode considering the satellite's modes of operation, orbital parameters and subsystem power usage.

In [11], researchers suggested a method for designing an EPS. This method involved analyzing the power budget of the satellite in consideration of its mission orbit and different modes of operation. To achieve this, they conducted a power budget analysis for a 1U CubeSat that took into account its mission orbit and various mission modes. Specifically, they estimated the power budget based on the communication mode, which they considered to be the mode that consumes the most power, according to their assessment.

The main contribution of this work is to evaluate the EPS design of a 3U Earth observation nanosatellite by examining its power budget. To achieve this, a methodology is developed that takes into account the orbit parameters and power consumption for the worst-case scenario. In the remaining sections, the mission description for the 3U nanosatellite is presented in section II. The orbit design, the operational modes and the power consumption of each subsystem are detailed in section III. Section IV investigates the sizing of the PV panels and the batteries used in this study. Section V presents the results and simulation. Conclusions are drawn in section VI.

II. Mission Description

The main objective is to design a 3U CubeSat for earth observation and scientific research that has a weight of 4.2 kg, and a size of 10 x 10 x 30 cm^3. This nanosatellite architecture is composed of the following subsystems: CDH which command and handle its data bus, GNC which control and maintain its attidute, COM (Rx & Tx) that ensure the communication with the Groud Station (GS), Thermal control (heaters), EPS and the payload, as shown in Figure 1. The latter is composed of a RGB camera that will be able to take high-resolution images of the Earth's surface and a Payload Controller (PC) that will analyze data, and transmit it through the COM(Tx) to the GS, that is located in Rabat.

This nanosatellite is designed for SSO orbit, with an altitude between 500-700 km, as it is recommended by the International Space Station (ISS). It has two pointing modes:

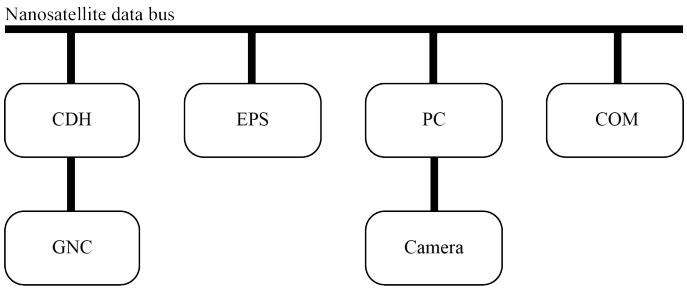

Fig. 1. 3U Nanosatellite architecture.

TABLE I
Orbital Parameters

Altitude h(km)	Inclination(°)	Orbits/day
500	97.4	15

- Sun alignment pointing attitude is the case where the side with the maximum PV cells is pointing to the sun, as seen in Figure 2.
- As shown in Figure 3, a nadir alignment with sun constraint pointing attitude occurs when the side +Y points to the center of the earth.

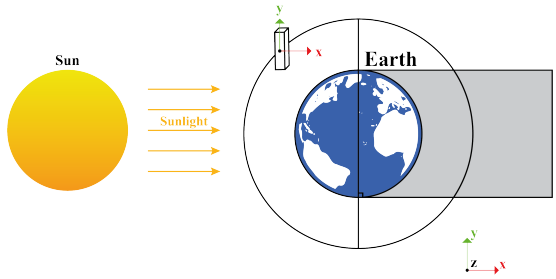

Fig. 2. -X Sun tracking pointing.

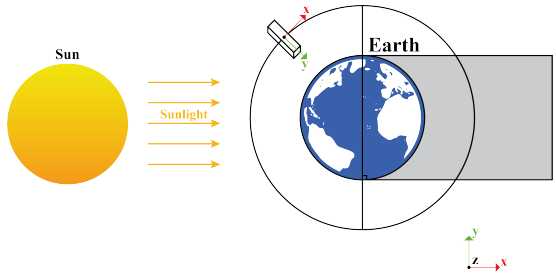

Fig. 3. +Y Nadir pointing with -X sun constraint.

The mission of this nanosatellite is to image Morocco during its lifetime. For this purpose, a coverage simulation is done using the STK software based on the selected orbital parameters shown in Table I.

The aim of the simulation is to determine the time needed to cover all the regions of Morocco with nadir pointing attitude, as shown in Figure 4. With this orientation and by computing all the access time, the simulation results shows that the

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on March 11,2024 at 21:33:21 UTC from IEEE Xplore. Restrictions apply.

coverage can be completed in 54 weeks. Thus, this part of the mission will be accomplished before the satellite's end of life.

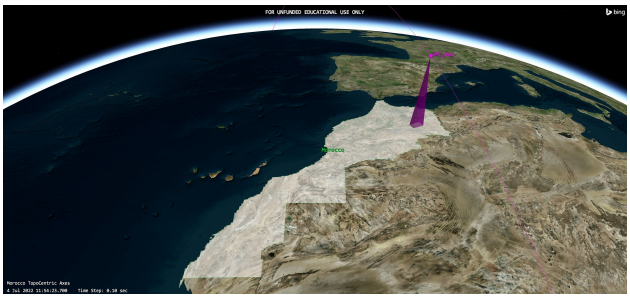

Fig. 4. Coverage over Morocco.

III. Methodology

A. Orbit consideration

The EPS design requires the knowledge of the following parameters:

- The orbital period T_o;
- The eclipse periode T_e;
- The sunlight period T_s.

The parameter T_o is determined by Kepler's Third Law [12] in the following equation:

$$T_o = 2\pi\sqrt{\frac{(R_e + h)^3}{\mu}} \tag{1}$$

where R_e is the radius of the earth equal to 6378 km, h is the altitude, and μ is the earth's standard gravitational parameter equal to 398601 km^3/s.

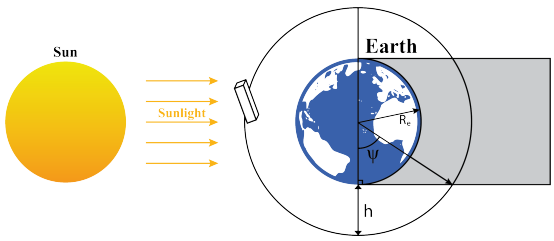

Fig. 5. Eclipse's calculation geometry.

Using the geometry illustrated in Figure 5, the T_e and T_s can be calculated using the following equations [13]:

$$T_e = (\frac{1}{2} - \frac{\arccos(\psi)}{\pi})T_o \tag{2}$$

$$T_s = (\frac{1}{2} + \frac{\arccos(\psi)}{\pi})T_o \tag{3}$$

where the angle ψ is defined by (4)

$$\psi = \frac{R_e}{R_e + h} \tag{4}$$

It can be noticed that the eclipse begins at ψ and ends at $\pi - \psi$.

TABLE II
Power consumption of each subsystem

Subsystems		Power(W)
EPS		0.3
CDH		0.4
GNC		0.8
Heaters		0.2
COM	Rx	0.5
	Tx	5
PC		0.3
Camera		3

B. Operational modes

The mission profile is defined using the following operational modes:

(1) Idle mode: The satellite is in the nominal mode but is not taking images or communicating with the GS

(2) Imaging mode: When the satellite takes images during daylight intervals.

(3) Transmission mode: When the satellite communicates with the GS during all the access.

(4) Transmission and Imaging mode: This mode combines (2) and (3), which means that the power consumption is high.

(5) Safe mode: When the batteries state of charge falls below 40%, the EPS's MCU enters the safe mode.

C. Power consumption

The power consumption of the nanosatellite's subsystems depends on their size, the components used and the mission requirements. After defining the mission's requirements, the power consumption of each subsystem is presented in Table II.

IV. EPS sizing

A. EPS description

The main function of an EPS for a nanosatellite is to provide reliable power to the satellite's payload and other onboard systems. The EPS typically consists of PV panels, batteries, a power management and distribution unit (PMAD). The solar panels provide power from the sun, the batteries store energy, the PMAD manages and distributes the power to the different subsystems onboard the satellite.

When designing the nanosatellite's EPS, the power budget must be established to size the PV panels and the batteries based on the power consumption during the sunlight and the eclipse.

In this work, the EPS architecture used and its state depending on the sunlight or eclipse period are shown in Figure 6. During the sunlight time, the PV panels provide the power for the subsystems and also the power to charge the batteries, except when the power needed exceeds the power produced by the PV panels. In this case, the subsystems will draw energy from the batteries, as seen in Figure 6(a). During the eclipse time, the battery pack is responsible for providing the necessary power for the subsystems as the PV panels are not operational in this period, as shown in Figure 6(b). Furthermore, the EPS will switch to the safe mode when the

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on March 11,2024 at 21:33:21 UTC from IEEE Xplore. Restrictions apply.

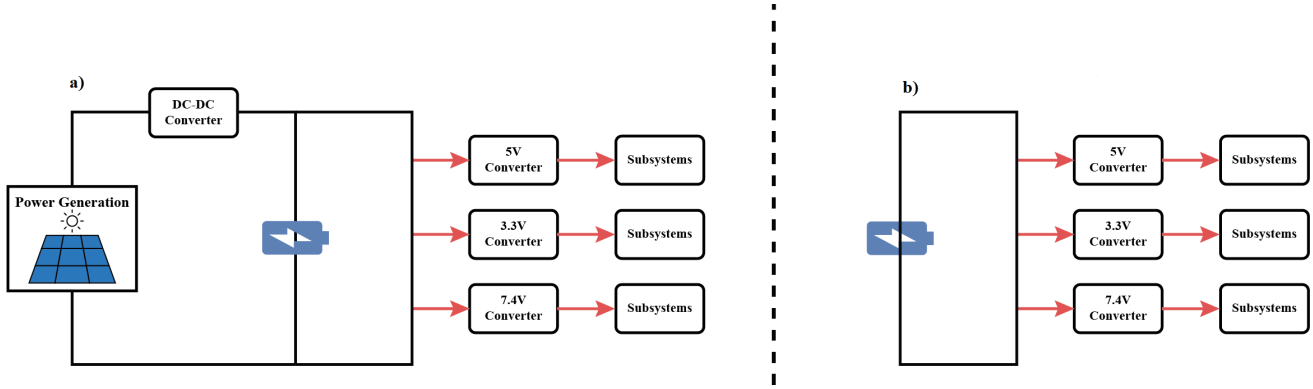

Fig. 6. EPS circuit states. (a) Sunlight circuit. (b) Eclipse circuit.

battery voltage drops below the threshold fixed to protect it from the discharging.

B. PV panel

The power generation of the nanosatellite is typically calculated by determining the total power required by its subsystems and its payload during the sulight period. Thus, the power generation required for this mission can be determined by considering the specific mission objectives, as described in section II.

The minimum power generation needed can be calculated using the following equation:

$$P_{pv,min} = \frac{\frac{E_s}{X_s} + \frac{E_e}{X_e}}{T_s} \tag{5}$$

where E_s, E_e, X_s and X_e are the energy consumed during the sunlight, the energy consumed during the eclipse, the efficiency from the PV panels to the loads and the efficiency from the PV panels to the batteries then the loads, respectively. When determining the values of X_s and X_e, the worst case is taken into account; in this study, X_s = 0.8 and X_e = 0.6.

C. Battery

The battery design for the nanosatellite should be done considering its size, weight, mission lifetime, power requirements during eclipses, and operating temperature. The mission's duration is estimated to be 3 years, which means that the depth of discharge (DoD) should not exceed 10%. The energy required for this mission can be calculated using the following equation:

$$E_{batt,req} = \frac{E_{ce}}{DoD}(1 + \frac{K}{100}) \tag{6}$$

where the E_{ce} is the energy consumed by the nanosatellite during the eclipse time and the K is the margin of 30% that compensates for the battery degradation.

V. Results & Simulation

For the studied nanosatellite, the SSO orbit with an altitude h = 500 km and inclination of 97.4° is chosen. The duration of one orbit T_o= 92.6 min, the eclipse period T_e= 36 min and the sunlight period T_s= 56.6 min are obtained using the eq. (1), eq. (2) and eq. (3), respectively.

TABLE III
INPUT PARAMETERS FOR THE POWER BUDGET SIMULATION

Parameter	Value
Altitude [km]	500
Start Time Simulation	4-July-2022 9:00:00 AM
Stop Time Simulation	5-July-2022 9:00:00 AM
PV cells for -X side [number]	6
Temperature[°C]	[-17.83, 67.57]
Maximum DoD[%]	10

TABLE IV
ELECTRICAL CHARACTERISTICS OF THE SOLAR CELL 3G30A

	V_{oc}(mV)	I_{sc}(mA)	V_{mp}(mV)	I_{mp}(mA)	Eff.(%)
BOL	2690	519.6	2409	502.9	29.6
EOL	2512	501.3	2244	485.1	26.3

This nanosatellite has body-mounted PV panels on its four lateral sides. The -X side is made up of six triple junction GaAs solar cells from AzurSpace with a 29.5% efficiency. Its electrical characteristics at temperature 28°C and solar irradiance of 1367 W/m^2 are summarized in Table IV.

Based on the table IV, the maximum power produced by the solar panel at the end of life (EOL) at 28°C and 1367 W/m^2 is 1.08 W/cell, which gives 6.53 W/panel.

The STK software is used to calculate the power generated in each orbit for one day using the following equation:

$$P_G = \sum_{n=1}^{4} P_n \tag{7}$$

where

$$P_n = S_c \eta A_n \cos(\alpha) \tag{8}$$

where S_c is the solar flux radiation, η is the efficiency of the PV panel, α is the incident angle of the sun with the PV panel, and A_n is the area of the PV panel on side n.

The worst case is considered in this simulation. The S_c is assumed to be constant, equal to 1316 W/m^2, the η=26.6%, the maximum area in the -X side A_1= 181.08 cm^2 and the

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on March 11,2024 at 21:33:21 UTC from IEEE Xplore. Restrictions apply.

temperature profile that varies between -17.83°C and 67.57°C, as illustrated in Figure 7.

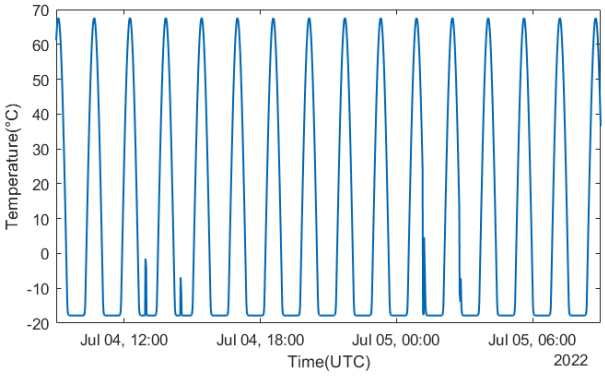

Fig. 7. Temperature variation.

TABLE V
EO MISSION POWER BUDGET

Subsystems		Power(W)	Ds(%)	Ps (W)	De%	P$_e$ (W)
EPS		0.3	100	0.3	100	0.3
CDH		0.4	100	0.4	100	0.4
GNC		0.8	100	0.8	100	0.8
Heaters		0.2	0	0	35	0.06
COM	Rx	0.5	100	0.5	100	0.5
	Tx	5	10	0.5	10	0.5
PC		0.3	20	0.06	0	0
Camera		3	10	0.3	0	0
Total Power(W)				2.7		2.36

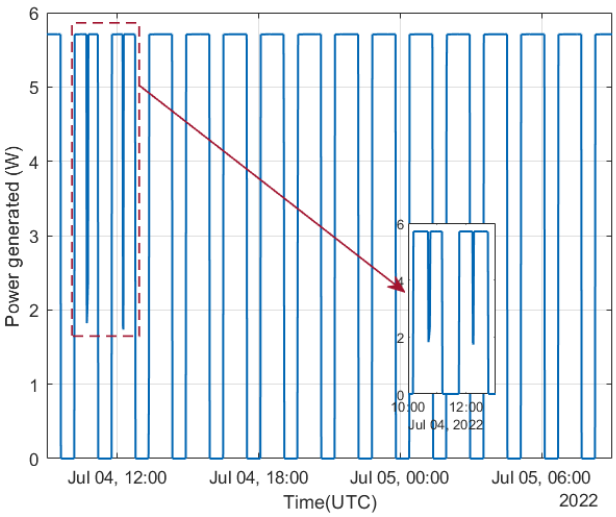

Fig. 8. Power generated during one day.

The maximum power generated by the PV panel for the side -X in its EOL with the sun's alignment attitude is P_n = 5.7 W, as shown in Figure 8. The power generated drops from 5.7 W to 1.82 W at 10:42 and from 5.7 W to 1.73 W at 12:14. Such changes can happen during the imaging mode, which means the nanostaellite changes its orientation from sun alignment to nadir alignment with sun constraint pointing, to take images over morocco.

Table V presents the power budget summary taking into account the following assumptions:

- During daylight access time, the camera captures images.
- EPS, GNC and CDH are always active.
- COM (Rx) is always ON.
- COM (TX) is ON during the GS visibility.

Furthermore, it is the worst-case scenario when the nanosatellite has to take images and transmit the telemetry to the GS at the same time. Figure 9 depicts the power consumption of the nanosatellite in mode (4). Figure 10 illustrates the available power generated for each orbit during one day.

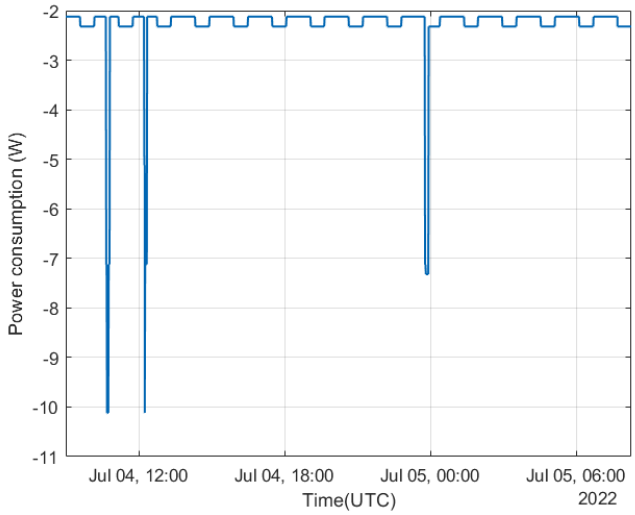

Fig. 9. Power consumption during one day.

It can be noticed that the power needed exceeds the power generated at 10.42 and at 12:14 because the COM Tx and the camera work at the same time during the sunlight period. Thus, the subsystems will draw power from the batteries.

Based on the battery sizing defined in Equation (6), the Li-Ion battery with an energy of 23 Wh from Nanoavionics is chosen. Examining the DoD performance in Figure 11, shows that the DoD value is limited to 8.5 % which confirms that the charge and the discharge of the battery are controlled correctly.

VI. CONCLUSION& FUTURE WORKS

The success of a nanosatellite's mission depends on a well-designed EPS, including the power generation as well as the power storage. The sizing of the PV panels and the battery pack are presented. The power budget of a 3U EO nanosatellite is evaluated using the instantaneous varation of the DoD of a SSO orbit. Based on the obtained results, it is safe to conclude that the chosen PV panels and the battery pack chosen are adequate for the mission specification presented.

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on March 11,2024 at 21:33:21 UTC from IEEE Xplore. Restrictions apply.

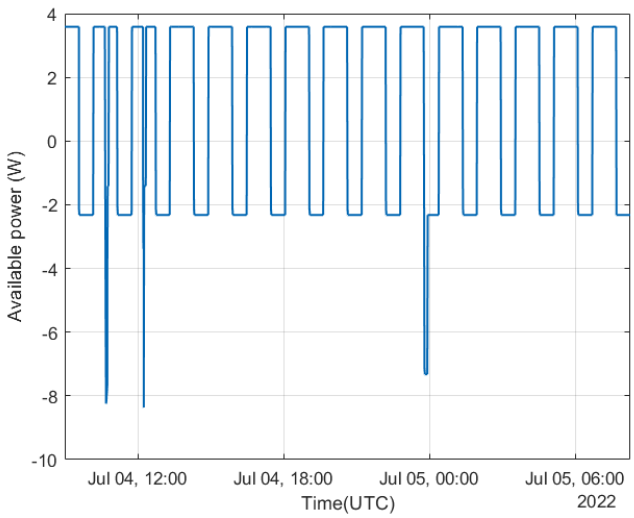

Fig. 10. Available power during one day.

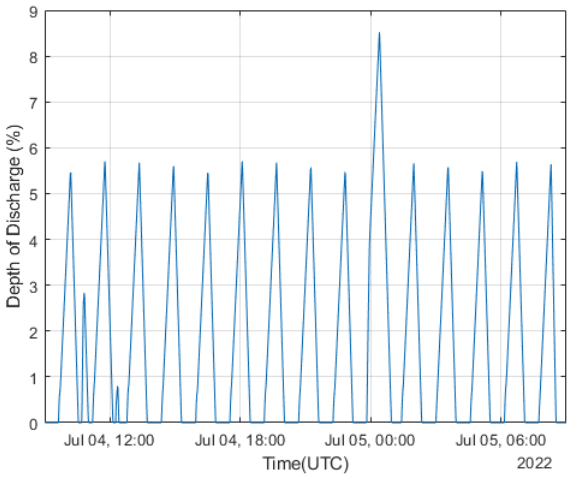

Fig. 11. Depth of discharge performance for each orbit during one day.

Future research will describe attitude control strategy and its performance, the expected accuracy, and assessing its potential impact on EPS. Furthermore, the modeling and the design of the EPS considering its efficiency and its reliability will be investigated .

ACKNOWLEDGMENT

This work was carried out in the frame of cooperation between the Royal Center for Space Studies and Research (CRERS) and the Mohammed V University in Rabat (UM5R).

REFERENCES

[1] I. Nason, J. Puig-Suari, and R. Twiggs, "Development of a family of picosatellite deployers based on the cubesat standard," in *Proceedings, IEEE Aerospace Conference*, vol. 1. IEEE, 2002, pp. 1–1.

[2] Y. Borthomieu, "Satellite lithium-ion batteries," in *Lithium-ion batteries*. Elsevier, 2014, pp. 311–344.

[3] A. Edpuganti, V. Khadkikar, M. S. El Moursi, H. Zeineldin, N. Al-Sayari, and K. Al Hosani, "A comprehensive review on cubesat electrical power system architectures," *IEEE Transactions on Power Electronics*, vol. 37, no. 3, pp. 3161–3177, 2021.

[4] A. Poghosyan and A. Golkar, "Cubesat evolution: Analyzing cubesat capabilities for conducting science missions," *Progress in Aerospace Sciences*, vol. 88, pp. 59–83, 2017.

[5] A. Daghouri, I. El Wafi, S. Elhani, M. Haloua, and Z. Guennoun, "Lqr and smc control design of a dc-dc converter based on kalman filter observer for a nanosatellite's eps: A comparative study," in *IECON 2022–48th Annual Conference of the IEEE Industrial Electronics Society*. IEEE, 2022, pp. 1–6.

[6] J. N. Pelton and S. Madry, "Power systems for small satellites," *Handbook of Small Satellites: Technology, Design, Manufacture, Applications, Economics and Regulation*, pp. 189–202, 2020.

[7] S. Dahbi, A. Aziz, S. Zouggar, M. El Hafyani, O. Morocco, A. Hanafi, M. Karim, I. Latachi, and T. Rachidi, "Power budget analysis for a leo polar orbiting nano-satellite," in *2017 International Conference on Advanced Technologies for Signal and Image Processing (ATSIP)*. IEEE, 2017, pp. 1–6.

[8] S. Rossi, A. Ivanov, M. Richards, and V. Gass, "The swisscube's technologies results after four years of flight," in *Proceedings of the International Astronautical Conference, Bejing, China*, 2013, pp. 23–27.

[9] P. LOPES, "Preliminary design and thermal study of the igosat project," Ph.D. dissertation, Master's thesis, Observatoire de Paris-Laboratoire d'Etudes Spatiales et d ....

[10] J. Pisane, "Oufti-1, the first student nanosatellite of belgium," in *Odissea award*, 2008.

[11] T.-Y. Park, B.-G. Chae, and H.-U. Oh, "Development of electrical power subsystem of cube satellite step cube lab for verification of space-relevant technologies," *International Journal of Aerospace System Engineering*, vol. 3, no. 2, pp. 31–37, 2016.

[12] M. Oredsson, "Electrical power system for the cubestar nanosatellite," Master's thesis, 2010.

[13] S. RM, "Computation of eclipse time for low-earth orbiting small satellites," *International Journal of Aviation, Aeronautics, and Aerospace*, vol. 6, no. 5, p. 15, 2019.

Authorized licensed use limited to: Universidad Tecnica Federico Santa Maria. Downloaded on March 11,2024 at 21:33:21 UTC from IEEE Xplore. Restrictions apply.